

PERSONALIZED LINUX & GIT CHEAT SHEET

Student No.: 201310836 | Quick reference for everyday command-line and Git work

BASIC LINUX COMMANDS

Command	Sample usage	My quick meaning
pwd	pwd	Prints the current directory path.
ls	ls -la	Lists files/folders; -a shows hidden items, -l gives details.
cd	cd project	Changes to another directory.
mkdir	mkdir project	Makes a new directory.
touch	touch index.html	Creates an empty file or updates its timestamp.
cp	cp a.txt b.txt	Copies a file; -r copies directories recursively.
mv	mv old.txt new.txt	Moves or renames a file/folder.
rm	rm file.txt	Deletes a file; -r can remove directories recursively.
cat	cat file.txt	Displays a file's contents in the terminal.
clear	clear	Clears the terminal screen.

BASIC GIT COMMANDS

Command	Sample usage	Description + useful flags
git init	git init	Creates a new Git repository in the current folder.
git status	git status -s	Shows current changes and staging state; -s = short output.
git add	git add .	Stages changes for the next commit; . stages all current changes.
git commit	git commit -m "Add page"	Records staged changes; -m supplies the commit message.
git log	git log --oneline	Shows commit history; --oneline gives a compact view.
git diff	git diff	Shows unstaged differences; --staged compares staged changes.
git restore	git restore file.html	Restores working-tree changes; --staged unstages a file.
git branch	git branch	Lists/manages branches; -a lists all, -d deletes a branch.
git checkout	git checkout main	Switches branches or restores files; -b creates a branch.
git switch	git switch main	Switches branches; -c creates and switches to a new branch.
git merge	git merge feature	Joins another branch into the current branch; --abort cancels a conflicted merge.
git tag	git tag v1.0	Creates a named version marker; -a creates an annotated tag.
git clone	git clone	Copies a remote repository to the computer; -b selects a branch.

GIT'S THREE-TREE ARCHITECTURE

WORKING TREE	STAGING AREA (INDEX)	REPOSITORY / HEAD
The files I edit on my computer. Changes here are not yet staged.	The selected snapshot prepared for the next commit. git add moves changes here.	The saved project history. git commit records the staged snapshot.

MY GIT WORKFLOW: Edit → git status → git add . → git commit -m "message" → git log --oneline → git push

QUICK REMINDERS

Before committing	Use git status to check what changed and what is staged.
Undoing work	git restore can discard working-tree changes; be careful because discarded changes may be difficult to recover.
Branches	Use git switch -c feature-name to create and move to a new branch.
Remote work	git clone copies a repository; git push sends local commits to a remote repository.

REFERENCES

1. Git Project. *Git Reference Documentation*. <https://git-scm.com/docs>
2. Git Project. *Git User Manual*. <https://git-scm.com/docs/user-manual>
3. GNU Project / Free Software Foundation. *GNU Coreutils Manual*. <https://www.gnu.org/software/coreutils/manual/>

The author used ChatGPT to assist with researching, organizing, and drafting the content of this cheat sheet. The author reviewed and edited the content as needed and takes full responsibility for the final content of the work.